Alpha-to-ion heat conversion: a damping–bandwidth obstruction

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Outcome and scope

The attempted capability is useful transfer of fusion-alpha energy into fuel ions, with sufficient power, accessible geometry and a better total energy balance than existing alpha-channeling approaches. That capability has **not** been constructed. The work below supplies an exact theorem for a reduced kinetic model, an exact finite-Hermite energy identity, and numerical tests of one homogeneous linear conversion stage. It does not establish a new instability, a reactor operating point, a nonlinear extraction fraction, historical novelty or experimental success.

Microscopic thermal memory was set aside after the preserved two-body mediator failed its composed logical-degeneracy test. Existing higher-dimensional passive models and active fault tolerance remain stronger references than that failed construction. This is a change of physical operation, not an extension of it.

The ingredient is Qin and Fisch, arXiv:2609.07732v1, Sections II and VI–IX: a minority-ion cyclotron pole can destabilize a thermal ion Bernstein mode. Its example transfers energy into coherent proton motion; the authors explicitly leave irreversible heating and practical thresholds open. Our question is whether adding a conserving relaxation channel closes that gap without losing the source instability to Doppler broadening.

Exact damping–bandwidth theorem for the reduced equation

Use frequencies relative to the energetic-ion harmonic. Let $g, d, K > 0$, with g a coupling frequency, d the damping rate of the positive-energy thermal mode, K the Gaussian parallel Doppler width, and $\Delta \in \mathbb{R}$ a freely adjustable detuning. Consider precisely

$$\delta - \Delta + id - \frac{g^2}{K} Z(\delta/K) = 0, \tag{1}$$

where, for $\operatorname{Im} z \geq 0$,

$$Z(z) = i \int_0^\infty e^{-t^2/4 + izt}\, dt.$$

This is a local normal form, not the full dielectric. It retains one positive residue, a constant thermal slope/damping, and a Gaussian parallel source. In particular it omits the source's parallel-gradient numerator and other poles.

Theorem. Some real detuning permits a root with positive growth if and only if

$$K < K_{\text{crit}} := \frac{\sqrt{\pi} g^2}{d}. \tag{2}$$

The greatest possible growth over all real detunings is attained at $\Delta = 0$ and is the unique positive solution of

$$\gamma + d = \frac{\sqrt{\pi} g^2}{K} \operatorname{erfcx}(\gamma/K). \tag{3}$$

Proof. Write $\delta = x + i\gamma$, $\gamma > 0$. The integral gives

$$\operatorname{Im} Z(x/K + i\gamma/K) = \int_0^\infty e^{-t^2/4 - \gamma t/K} \cos(xt/K)\, dt \le \sqrt{\pi} \operatorname{erfcx}(\gamma/K) < \sqrt{\pi}.$$

Taking the imaginary part of (1) proves necessity of (2). At zero detuning, $\delta = i\gamma$ solves the real part identically. The right side of (3) strictly decreases with γ, while the left side increases. At zero growth their ordering is favorable exactly under (2); at infinite growth it is unfavorable. Continuity therefore gives a unique positive root. The preceding cosine bound shows that any root at any detuning has growth no larger than it. Equality requires $x = 0$, which in (1) requires $\Delta = 0$. This proves the claimed optimum and excludes an improvement from mere frequency retuning. At equality in (2) the centered root is marginal, not growing. □

In the cold-source limit, (3) becomes

$$\gamma(\gamma + d) = g^2, \qquad \gamma = \frac{2g^2}{d + \sqrt{d^2 + 4g^2}}. \tag{4}$$

For a strongly damped conversion stage, $d \gg g$, growth is approximately g^2/d, and the allowed Doppler width is only $\sqrt{\pi} g^2/d$. Efficient relaxation can therefore narrow the source's allowed frequency spread. This conclusion is an exact statement about (1). It is not a stability theorem for every alpha distribution, every plasma branch, or every geometry.

A conserving model separates stored wave energy from heat

Let f_1/f_0 have normalized three-dimensional probabilists' Hermite coefficients $a_{n_x n_y n_z}$, with Maxwellian f_0 of variance $\sigma_p^2 = T_p/m_p$. Frequencies below are in units of Ω_p. The linearized Dougherty model is

$$C = -\nu N + \nu \Pi_1 + 2\nu |u_E\rangle\langle u_E|, \quad u_E = (|200\rangle + |020\rangle + |002\rangle)/\sqrt{3}, \tag{5}$$

where N is total Hermite degree and Π_1 projects onto the three degree-one momenta. Thus density, all three momenta and total three-dimensional energy are null modes; every remaining eigenvalue is nonpositive. The model and its Hermite structure are established; see Anderson and O'Neil, *Eigenfunctions and Eigenvalues of the Dougherty Collision Operator*, Sections I–II. Its constant drag/diffusion coefficients do not reproduce the full velocity dependence of Coulomb collisions. We have not calibrated ν to a measured or full Landau collision operator.

Use the field convention $\exp(i\mathbf{k}\cdot\mathbf{x} - i\omega t)$, and normalize $q_p\phi/T_p = 1$. The finite Galerkin response obeys

$$(A - C - iwI)a = b, \quad b = -i\frac{\sigma_p}{\Omega_p}(k_\perp|100\rangle + k_\parallel|001\rangle),$$

where streaming plus gyromotion A is anti-Hermitian. Define

$$P = \frac{\omega_{pp}^2}{k^2\sigma_p^2}, \quad \chi_p = -Pa_{000}, \quad \gamma = \operatorname{Im} w > 0.$$

Multiplying the response equation by $a^\dagger$, taking its real part, and using its density row gives the **exact finite-matrix identity**

$$\frac{\operatorname{Re}[-iw\chi_p]}{\gamma} = P\|a\|^2 - \frac{P}{\gamma}a^\dagger Ca. \tag{6}$$

The first term is reversible fluctuation free energy; the second is nonnegative quadratic entropy production. In the specified relaxation model it is the irreversible conversion of fluctuation free energy into background heat. First-order conservation of total energy in (5) is consistent with this second-order relaxation. It does not justify equating all coherent proton power with heat. Nor does (6) certify nonlinear heating in a real device.

For $k_\parallel = 0$, the full 3D operator leaves $n_z \in \{0, 2\}$ invariant; both are retained. At finite parallel wavenumber, extra n_z values are included and their cutoff is checked. An initial 2D trial was replaced by the full 3D model; only the latter supports the physical numbers below.

Specified full-dielectric test

The reference uses $B = 0.05$ T, $n_p = 10^{17}\,\mathrm{m}^{-3}$, $n_\alpha/n_p = 0.02$, $T_p = 5$ eV, $T_e = 10$ eV, alpha mass $4m_p$, and mean alpha energy 24 keV. These are a cold laboratory-scale reference, not a proton–boron fusion plasma. At finite $k_\parallel$, electrons and alphas have their full collisionless kinetic susceptibilities; protons use (5).

For independently normalized gyrotropic distributions, the kinetic integral is

$$\chi_s = \frac{\omega_{ps}^2}{k^2}\sum_n \int d^3v \frac{J_n^2(k_\perp v_\perp/\Omega_s)}{\omega - k_\parallel v_\parallel - n\Omega_s}\left(\frac{n\Omega_s}{v_\perp}\partial_{v_\perp} f_s + k_\parallel \partial_{v_\parallel} f_s\right). \tag{7}$$

Integrating its Gaussian parallel factor gives Z and its derivative. Large-argument upper-half-plane expansions avoid cancellation. Cold-ring responses are averaged over an explicitly normalized perpendicular speed distribution, a Gaussian in $u = v/v_0$ of width 0.1, truncated at six standard deviations and rescaled to keep the stated mean energy. This is an imposed test population, not an acquired fusion birth or loss-cone distribution. With parallel temperature $T_{\alpha\parallel}$, mean perpendicular energy is $24000 - T_{\alpha\parallel}/2$ eV, so total energy is unchanged.

Benchmark and perpendicular-width correction

At $k_\perp = 251.74\,\mathrm{m}^{-1}$, the source's cold benchmark is reproduced:

$$w = 1.49837407794 + 0.00133377037i,$$

with proton, alpha and electron powers in units $2\gamma W_E$ equal to 11843.15987, −11848.43978 and 4.2799083. Including the growing field closes the balance. Here $W_E = \epsilon_0|E|^2/4$.

The actual thermal-only crossing at $w = 1.5$ is $k_* = 250.147719979\,\mathrm{m}^{-1}$, $b_p = 1.306503224$. For the mass/response conventions just stated, the *cold* third-alpha residue there is negative, −0.07217383435 in dimensionless w^2 coordinates. The positive-residue theorem therefore does not certify that nominal cold crossing. The source's Section VIII already says this crossing needs revalidation; this discrepancy does not refute the conditional root-pole theorem.

Finite perpendicular width changes the sign favorably in this example. At the same crossing, widths 0.02, 0.1 and 0.2 give residues 0.0006146618, 0.0006671396 and 0.0008915813; corresponding collisionless growth rates are approximately 0.0001606113, 0.0001673271 and 0.0001934365. Broadening need not destroy the perpendicular drive because it averages an oscillatory Bessel derivative.

Harmonic truncation 190 versus 450 and radial quadrature 160, 320 and 640 were compared. Width 0.1 is converged for the reported quantities. An initial width-0.4 residue was aliased; its value was withdrawn. The converged width-0.4 residue is −0.03979934, with no validated growing continuation branch found. Failure to find one is not a global stability proof.

Relaxation helps heat conversion while suppressing growth

For width 0.1, $k_\parallel = 0$ and $k_\perp = k_*$, numerical solutions of the full dielectric give:

model collision rate ν/Ω_p	growth γ/Ω_p	irreversible proton heat / alpha power loss
0	0.0001673271	0
0.000001	0.0001653220	0.02382
0.00001	0.0001483660	0.21379
0.0001	0.0000603649	0.86985
0.001	0.0000069280	0.998286
0.003	0.0000023135	0.999809

These are *fractions of the infinitesimal eigenmode's instantaneous alpha power loss*. They are not fractions of all alpha birth energy recovered, and they do not determine absolute power without amplitude and saturation. Rate 0.001 was chosen as a model stress test, not supplied as a calibrated physical service.

The extracted local parameters are $g = 0.0001673270\Omega_p$ and $d = 0.004034533\Omega_p$, giving $K_{\mathrm{crit}} = 1.230025 \times 10^{-5}\Omega_p$ and cold growth $6.927778 \times 10^{-6}\Omega_p$. This closely describes the full numerical branch. Agreement is an asymptotic/numerical check; it does not promote the exact theorem about (1) into a full-dielectric theorem.

Direct marginal solve and retuning

For $\nu/\Omega_p = 0.001$, solve both real and imaginary dielectric equations on the real-frequency axis rather than treating failed unstable-root searches as stability. At fixed $k_\perp = k_*$, the results are:

$T_{\alpha\parallel}$, eV	marginal $k_\parallel$, m^{-1}	$\pi/k_\parallel$, m
1	0.00880865	356.65
10	0.00269373	1166.26
100	0.0008539486	3678.90
1000	0.000277841	11307.15

For 100 eV, Hermite total orders 20, 24 and 28 with parallel cutoffs 3, 4 and 5 give marginal wavenumbers 0.000853948701, 0.000853948616 and 0.000853948615. Marginal residuals are below 1.1×10^{-10}. Tracking the same branch at 0.8 and 1.2 times this wavenumber gives respectively positive growth $2.0807632 \times 10^{-6}\Omega_p$ and negative growth $-2.2682245 \times 10^{-6}\Omega_p$.

Retuning $k_\perp$ to place the marginal wave exactly at the alpha harmonic gives $k_\perp = 250.144493933\,\mathrm{m}^{-1}$ and $k_\parallel = 0.000853982666\,\mathrm{m}^{-1}$. Thus this resonance retuning does not remove the small bandwidth. It is not an optimization over every mode, distribution, collision rate or geometry. The parallel-gradient source term is included in these numerical tests, which is why the reduced theorem is not an exact identity for the full calculation.

For a uniform column with Dirichlet potential at both ends, the smallest nonzero parallel wavenumber is π/L. This one stage therefore requires roughly 3.7 km at 100 eV. Periodic/closed-field/flute geometries can admit exactly $k_\parallel = 0$; the length inference does not apply to them. We have not acquired a finite-device eigenmode or proved a no-go theorem for alpha channeling.

Same-resource comparator and unclosed capability gates

At identical $f_0, k, \omega, \phi, \nu$, an externally maintained wave has precisely the same linear proton response a and heat production as this self-excited field. Equation (6) gives, without dividing by growth,

$$Q_p = -2\Omega_p W_E\, P\, a^\dagger C a.$$

Thus the constructed relaxation operator supplies no new heating advantage at a fixed field. A practical advantage would have to come from affordable alpha extraction and reduced external drive after all losses and source preparation are charged. Neither absolute extraction power nor such a total-cost advantage has been demonstrated.

The stronger established architecture couples spatial extraction with energy diffusion and selective ion damping. Fisch and Herrmann's 1999 tutorial, Section 2, explains why the energy/position paths, wave competition with slowing down, and multiple-wave spectrum matter for a full birth distribution. The present homogeneous ring stage has not beaten that architecture with equal wave/control resources. No completed reactor-level comparison is claimed.

The decisive missing operations are an earned energetic-ion distribution and its replenishment cost; a finite-device mode that survives parallel/field inhomogeneity and escape; a calibrated conserving multispecies collision model; nonlinear extraction, saturation and ash removal; and a full equal-resource energy balance including electron heating, radiation, drive and confinement. Adding a positive growth rate or a high instantaneous heat fraction does not supply these operations. The source's own single-wave plateau warning is also not an acquired many-wave recovery scheme.

Evidence and limitations ledger

Private working directory: `scratch/astra-alpha-transfer-20261104` under `/workspace`. Numerical receipts are `DISPERSION_RESULTS.json`, `WIDTH_RESULTS.json`, `QUADRATURE_CHECK.json`, `PARALLEL_GATE.json`, `MARGINAL_RESULTS.json` and `NORMAL_FORM_RESULTS.json`. Source code and logs remain private; this proof archive contains no Python. Computation used Python 3.12.14, NumPy 2.3.5 and SciPy 1.17.0 on the provided CPU environment, with no external compute purchase. The repository ingredient checkout is `3e6f6fa576181275bb8ed28164e7cacd51055789`.

The finite-Hermite null space, anti-Hermitian streaming/gyromotion, identity (6), full field/species power balance, quadrature and selected marginal convergence were checked. Near-pole cancellation made an absolute power-balance tolerance inappropriate; it was replaced by a relative

tolerance, with the dielectric residual retained. Numerical solves are floating-point diagnostics, not interval certificates or a global root census. The only exact stability statement here is the explicit reduced-model theorem (1)–(3).

The current Astra N1639 packet was consulted as a warning about missing RF-to- heat accounting. Its reduced zonal-flow primitive is a different mechanism and supplies no theorem used here. Current packet access did not supply a gyrokinetic heat advantage or the missing energetic-ion kernel. No held archive was fetched.

Access limits: two OSTI primary endpoints returned HTTP 502 during this pass; their full text was not acquired. An APS search result for the 2022 proton–boron power-balance paper required authorization; no alternate full-text route was used. These are literature-comparison limits, not physical evidence against any method. Other cited primary sources were read through the web tool. This record preserves a substantial scoped obstruction and adverse numerical tests; the civilisation-changing objective remains open.